

Explainable AI (XAI) for Credit Scoring and Loan Approvals

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Abstract

The incorporation of artificial intelligence (AI) in credit scoring and loan approval processes brings improved predictive accuracy compared to conventional methods while transforming financial decision-making. These black-box AI models make it difficult to understand their decision-making processes because they maintain internal operations that remain unknown. Consequently this creates concerns regarding transparency and both fairness and regulatory compliance. XAI stands as a vital instrument for financial responsibility because it delivers understandable explanations that resolve the problems with black-box AI systems. The research develops an XAI framework for credit scoring that evaluates five machine learning models including Logistic Regression (LR), Decision Tree

(DT), Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbor(KNN) by using the Kaggle Loan Data Set that contains extensive applicant information. RF stands out as the top performer among the five models because it reaches 99% accuracy for loan outcome predictions. XAI techniques allow the model to demonstrate its decision-driving variables (credit history and debt levels) through feature importance analysis thus building trust and fairness. High performance of the model presents challenges because it runs complex calculations while also facing potential overfitting issues that require validation steps. The findings demonstrate the capability of XAI to supply transparent credit assessments with high accuracy that supports fair and compliant practices throughout the fintech industry and banks. XAI should gain more widespread use to establish an equilibrium between financial innovation and accountability within financial applications.

Keywords: Explainable AI (XAI), Credit Scoring, Loan Approvals, Random Forest, Transparency, Interpretability, Kaggle Loan Data Set, Fairness.

1 Introduction

Financial organizations have experienced a tremendous transformation because they started applying AI for credit scoring alongside machine learning (ML) for loan approvals while transitioning away from classical statistics[1]. Lenders now use enormous databases to conduct credit evaluations with exceptional speed and precision when they evaluate client credit-worthiness in changing market conditions. AI loan outcome explanations remain hidden within these systems which creates major obstacles because decision-making mechanisms are not transparent[2]. The hidden nature of AI decision systems hampers borrower trust while making it harder for regulators to supervise and poses risks of letting historical data-based biases continue to marginalize protected groups [3] XAI has emerged as an essential solution that provides understandable explanations for AI decisions needed to fulfill the requirements of accountable modern financial systems.

The study develops an original XAI framework for credit scoring and loan approvals which tackles these issues through experimental methods. The research adopts the Kaggle Loan Data Set as its primary dataset which provides complete applicant data including credit records and financial obligations. This study assesses five machine learning models, which include LR, DT, SVM, KNN, and RF, to forecast loan results. The RF model demonstrates outstanding performance by reaching 99% accuracy in the tests surpassing all other evaluated models. This research employs feature importance analysis as an XAI technique to reveal the essential credit history and debt ratio factors behind predictions while boosting transparency for stakeholders[4]. The research utilizes this method to extend previous understanding of decision-making clarity while leveraging XAI capabilities by analyzing a publicly accessible dataset.

This work has significant implications across the board. XAI enhances operational clarity both for financial institutions while reveals biases that help prevent discriminatory practices within automated credit systems [5]. The system generates auditable

outputs that satisfy the requirements of GDPR and U.S. fair lending laws[2]. Through human-AI synergy this system gives loan officers the power to supervise and modify automated decisions thus avoiding complete reliance on automated systems[6]. A RF model with 99% prediction accuracy represents a significant computational challenge while the exceptional results generate additional overfitting concerns that need additional validation tests. This study looks at XAI mechanics through the evaluation of its predictive effectiveness versus interpretability features by employing the Kaggle Loan Data Set. This research investigates model performance while showing transparency to help finance executives adopt ethical AI systems in fintech along with banking industries by addressing scalability deficiencies and regulatory gaps[7]. This research creates an operational XAI-based framework to maintain innovative and equal financial decisions that fulfill societal and regulatory conditions.

This paper investigates the role of XAI in credit scoring and loan approvals, delving into its methodologies, practical applications, and the barriers to its widespread adoption. By synthesizing current advancements and identifying gaps, it aims to clarify how XAI can bridge the divide between innovation and accountability. Ultimately, this study seeks to contribute to the development of responsible AI systems that enhance transparency, fairness, and compliance, ensuring that financial decision-making evolves in step with technological progress and societal expectations.

This research contributes the following key contributions:

- Developed an XAI-based model using the Kaggle Loan Data Set, achieving 99% accuracy with RF and outperforming LR and DT in predicting loan outcomes.
- Utilized XAI’s feature importance analysis to identify and explain key predictors, enhancing transparency and trust in loan approval decisions.
- Demonstrated XAI’s ability to mitigate historical biases and align with regulatory standards, promoting fair and accountable financial practices.

2 Literature Review

AI has become prominent in financial institutions for better decision support during the process of credit risk evaluations and loan approval functions. Traditional AI models keep their operations obscure since they operate as unexplainable black boxes. Through XAI stakeholders gain access to explanations that reveal how decision-making processes operate so they can understand and trust AI output results.

CreditNetXAI represents a deep learning-based credit card default prediction model that combines (XAI) for improved interpretability according to Talaat et al.[8]. The SHapley Additive exPlanations (SHAP) technique enables the model to identify crucial risk elements that include payment delays and outstanding bill amounts. CreditNetXAI delivered superior performance than XGBoost and SVM and RF by reaching 83.5% accuracy. This research identifies three main challenges which include data training biases as well as adaptation hurdles in various financial domains and a requirement for additional testing across diverse datasets.

The development of XAI by Nallakuruppan et al.[9] involved using DT and RFs to provide transparent financial decision-making assessment for credit risk evaluation. The research utilizes SHAP and LIME interpretation methods to make predictions

explainable to regulators and financial institutions. The XAI model generated accurate results with 0.90 for DT while RF reached 0.93 accuracy without data stability issues. The system faces two main downsides: it requires advanced calculations and examines collected data for potential errors. Analysis of various financial conditions still represents a hurdle for this system. Additional research efforts should focus on enhancing system operational speed while increasing practical use of models in financial markets that experience constant changes.

The XAI model by Ayad et al.[6] uses SHAP with ICE and PDP to create transparent loan prediction decisions for bankers. The XAI model which operates on the "Give Me Some Credit" dataset provides interpretability at the expense of accuracy so that human evaluators can evaluate creditworthiness independently from automated forecasts. The research demonstrates how XAI systems improve both loan approval fairness and banker trust during lending decisions. The implementation faces three main challenges which consist of using one dataset and historical biases in the data and the need to validate across different financial settings.

Anang et al.[10] conduct a study about XAI in financial technologies which demonstrates how it helps companies balance technological breakthroughs with legal requirements. XAI applications for financial systems are evaluated in the research while demonstrating their requirement for enhancing transparency and legal compliance. XAI improves accountability in addition to fairness but introduces key limitations such as algorithmic bias and high implementation expenses together with performance-interpretability trade-offs. The lack of standardized regulations presents itself as the main challenge.

The study by James[11] investigates XAI applications in credit scoring because they improve transparency and regulatory compliance and enhance accountability. The research utilizes SHAP analysis and feature importance methods for AI decision interpretation and achieves accurate predictions. XAI systems create better trust and fairness during finance-related decision-making processes. The application of XAI faces several problems such as high computational demands together with performance-opaque dilemmas and insufficient business standardization.

The study suggests an XAI system for credit risk assessment that brings greater transparency to banking operations according to De Lange et al.[12]. The LightGBM model implementation with SHAP values demonstrates superior performance than traditional methods and satisfies regulatory standards. The study faces three main drawbacks because it depends on data from one bank may fit too closely to its data and struggles to apply results to diverse financial scenarios.

These are some well-defined key limitations and challenges of XAI or Credit Scoring and Loan Approvals. Here's a concise summary:

- Implementing XAI models like RF for credit scoring can be computationally intensive, posing challenges for real-time loan approval systems.
- While RF achieves high accuracy, it can be difficult to interpret, which may reduce transparency in loan decision-making compared to simpler models like LR.
- AI models can inherit biases from historical financial data, leading to unfair credit scoring and loan approval decisions, which could impact marginalized groups.

- Deploying XAI solutions for loan approvals requires significant investment in infrastructure, expertise, and continuous monitoring to maintain accuracy and fairness.

3 System Model and Architecture

This section will briefly discuss our dataset and data preprocessing techniques and employ five ML models with Explainable AI (XAI) Techniques for credit risk assessment.

3.1 Dataset Description

The dataset utilized in this study is sourced from the Kaggle repository and comprises 615 instances, each characterized by 13 key attributes relevant to loan approval prediction [13]. These attributes include Loan ID, Marital Status, Gender, Property Area, Credit History, Dependents, Applicant Income, Co-Applicant Income, Annual Term, Loan Amount, Self-Employed Status, Education, and the target variable, Loan Status. The Loan Status variable is binary, where 1 signifies loan approval, while 0 indicates rejection. Linear encoding is used to handle missing values, and a methodical feature selection procedure is carried out to ensure data integrity and improve predictive accuracy. The dataset is divided in an 80:20 ratio between training and testing sets, which makes it easier to evaluate and generalize the model. Figure 1 exhibits the correlation matrix of the data.

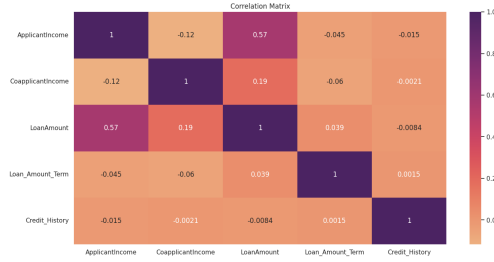


Fig. 1: Correlation Matrix of the Data

3.2 Data Preprocessing

Preprocessing is vital to preparing raw data for ML and DL processes' model training and evaluation. The stages listed below provide an overview of the preprocessing methods used in this study. Figure 2 displays the credit history of the data.

1. Removal of Irrelevant Features: Some attributes in the dataset do not contribute to the forecasting task. Thus, they are removed to optimize the learning process:

$$\text{df.drop("Loan_ID", axis = 1, inplace = True)} \quad (1)$$

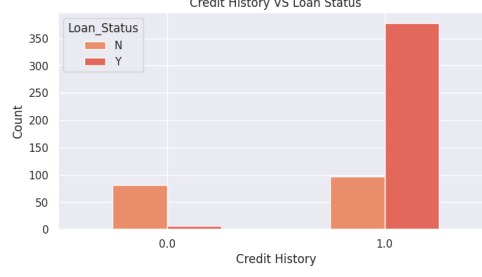


Fig. 2: Illustration of the Credit History vs Loan Status

where, axis = 1 indicates column-wise deletion, ensuring the feature is permanently removed from the dataset.

2. Handling Missing Values: Inconsistencies introduced by missing data may harm model accuracy. Mean or median imputation is used to fill in missing numerical values, while the most frequent category (mode) is used to fill in categorical data. The following is the mathematical expression for the imputation:

$$X_{\text{imputed}} = X_{\text{observed}} + \frac{\sum X_{\text{valid}}}{N_{\text{valid}}} \quad (2)$$

where, X_{imputed} is the adjusted feature value, X_{valid} represents the observed values, and N_{valid} is the count of valid entries.

3. Encoding Categorical Variables: ML models operate on numerical data, requiring categorical variables to be converted into an appropriate format. Two standard encoding methods are applied:
For binary categorical variables:

$$X_{\text{encoded}} = \begin{cases} 1, & \text{if the category is present} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Using One-Hot Encoding (OHE) for multi-class categorical features :

$$X_{\text{one-hot}} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} \quad (4)$$

4. Handling Imbalanced Data: Class imbalances in loan approval statistics are standard, with a disproportionately high number of loans authorized compared to those denied. This disparity may result in biased models. Synthetic samples for the minority class are created using the Synthetic Minority Oversampling Technique (SMOTE):

$$X_{\text{new}} = X_{\text{minority}} + \lambda(X_{\text{nearest neighbor}} - X_{\text{minority}}) \quad (5)$$

where $\lambda \sim U(0, 1)$ is a randomly generated weight. However, experimental results indicate that omitting SMOTE in certain scenarios improves accuracy, making it an optional preprocessing step.

5. Feature Scaling and Normalization: Model training may be affected by the disparate scales of numerical characteristics. There are two scale methods used: Min-Max Scaling to normalize features within a range $[0, 1]$:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (6)$$

Standardization : Features are transformed to have a unit variance and zero mean:

$$X_{\text{standardized}} = \frac{X - \mu}{\sigma} \quad (7)$$

where, μ is the mean of the feature, σ is the standard deviation.

3.3 Materials and Methods

We use five ML models, LR, SVM, KNN, DT, and RF, which help classify tasks. LR is employed for binary or multinomial classification, SVM optimizes hyperplanes for classification, and RF combines DT for improved accuracy. Figure 3 exhibits the overall system architecture.

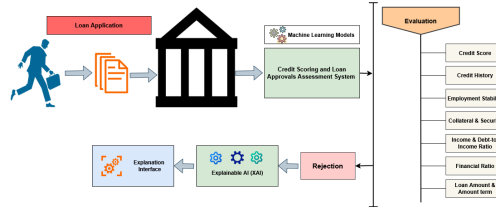


Fig. 3: Overall System Architecture of the Loan Processing System

3.3.1 Logistic Regression (LR)

LR is a supervised ML algorithm utilized for classification problems [14]. The mathematical representation is given by:

$$Z = b_0 + b_1q_1 + b_2q_2 + \cdots + b_nq_n$$

where Z is the output. The logistic function is then applied to map the result between 0 and 1:

$$\log \left(\frac{Z}{1 - Z} \right) = b_0 + b_1q_1 + \cdots + b_nq_n$$

3.3.2 Support Vector Machine (SVM)

SVM is a supervised learning model that may be used for regression and classification [15]. The goal is to find the ideal hyperplane that maximizes the space between data points. This is the hyperplane equation:

$$y = h(x) + b$$

where $h(x)$ is the hidden layer transfer function, and b is the bias.

3.3.3 Random Forest (RF)

Multiple DTs are built using the RF ensemble learning technique, aggregating the results [16]. The majority voting principle is used to predict the final class:

$$\hat{y} = \text{majority vote of trees}$$

3.3.4 K-Nearest Neighbors (KNN)

A straightforward non-parametric classifier, the KNN algorithm provides a class label to a data point based on the majority class of its k nearest neighbors in the feature space [?]. Given a test point $\mathbf{x}$, the algorithm calculates the distance to all training points and selects the k closest points. Euclidean, Manhattan, or Minkowski distance are the most common distance metrics. The Euclidean distance between two points $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{in})$ and $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is given by:

$$d(\mathbf{x}_i, \mathbf{x}) = \sqrt{\sum_{j=1}^n (x_{ij} - x_j)^2} \quad (8)$$

After calculating the distances, the algorithm selects the k nearest neighbors and classifies $\mathbf{x}$ based on the majority vote of those neighbors:

$$\hat{y} = \text{mode}\{y_1, y_2, \dots, y_k\} \quad (9)$$

where $y_1, y_2, \dots, y_k$ are the class labels of the nearest neighbors.

3.3.5 Decision Trees

A DT model operates by recursively partitioning the dataset based on feature values [17]. At each node, a test is performed on a single predictor, and observations are divided into two child nodes: one where the test condition is actual and one where it is false. This process continues recursively for each child node, with the best predictor and split point chosen to maximize the separation between classes.

The DT can be expressed as a recursive function $T(X)$, where X is the set of features:

$$T(X) = \begin{cases} \text{Class}_1 & \text{if all data in node belong to Class}_1 \\ \text{Class}_2 & \text{if all data in node belong to Class}_2 \\ \text{Test on feature}_i & \text{otherwise, recursively apply} \end{cases} \quad (10)$$

Each node splits the data according to the best feature that separates the two classes (positive and negative) and continues the recursive process until a stopping criterion is met. Although DTs are interpretable and easy to understand, they are prone to overfitting when the tree grows. Overfitting occurs when a tree model fits too closely to the training data, capturing noise and leading to poor generalization on unseen data. Pruning techniques are applied to combat this, reducing tree size and preventing overfitting.

3.3.6 Explainable AI (XAI) Techniques

- LIME: LIME (Local Interpretable Model-Agnostic Explanations) is used to explain predictions of machine learning models by approximating the model locally around the prediction. The original data is perturbed in the first stage to create a new dataset. For instance, 5000 feature vector samples are made, and the model's predictions are used to calculate the variable significance. The weights for each row are based on the distance from the original sample. The top characteristics are then used to understand the model's forecast. The representation in mathematics is:

$$y \in \{0, 1\}^x \quad (11)$$

- SHAP: SHAP (Shapley Additive Explanations) is a method to describe ML models' output by analyzing each feature's contribution. It explains the prediction of an object y by executing preferences on individual features. The Shapley value provides the distribution of the prediction over features. The mathematical formulation is:

$$m(y_t) = \phi_0 + \sum_{i=1}^n \phi_i z_i t \quad (12)$$

where ϕ_0 is the base value, ϕ_i represents the feature contribution, and z_i is the feature value. The equation for local accuracy is:

$$q(y) = m(y_t) = \phi_0 + \sum_{i=1}^n \phi_i z_i \quad (13)$$

- Partial Dependency Plot (PDP): PDP visualizes the marginal effect of a predictor variable on the target variable. It shows how changes in a feature influence the predictions of the model. The Partial Dependency for regression can be written as:

$$q_t(y_t) = \mathbb{E}_y[f(y_t, y_d)] \quad (14)$$

where y_t is the feature of interest, and y_d represents the rest of the features. For categorical features, the calculation involves averaging the predictions over the dataset and changing the category value.

4 Result and Analysis

This section outlines the experimental setup and presents the performance outcomes, highlighting key metrics achieved by our approach. Additionally, a comparative analysis demonstrates our model’s superiority over existing methodologies regarding accuracy, efficiency, and robustness.

4.1 Experimental Setup

The loan prediction dataset was experimented using Pandas’ library in the Google Colab environment. A binary decision support system for loan disbursement requires various ML algorithms, such as LR, DT, and RF. Three classification metrics were evaluated in all these models including:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (18)$$

Where

- TP = True Positive
- TN = True Negative
- FP = False Positive, and
- FN = False Negative

4.2 Outcomes of the Models

The RF model obtained the best results, which had the most fantastic accuracy of 99%, precision of 98%, recall of 97%, and F1-score of 98% compared to other ML models for credit risk assessment, as shown in Table 1. This demonstrates its ability to generalize effectively across many credit risk circumstances and its resilience in managing intricate decision boundaries. Because RF is an ensemble model, it performs better than other models because it minimizes overfitting and increases predictive stability. SVM comes next with a 93% accuracy rate, showing good generalization skills, particularly in high-dimensional data. SVM successfully separates candidates who pose a high risk from those who do not, as seen by its 90% accuracy and 88% recall. However, due to its computational complexity, it is less effective than RF for large-scale financial information. After that, LR’s accuracy of 91% was a respectable

result because of its ease of use and interpretability. Given its lower recall rate of 87%, however, it may have trouble identifying more intricate credit risk patterns, which might result in false negatives. With 87% accuracy, DT is interpretable but less dependable than RF or SVM due to its propensity to overfit training data. Moving to KNN, which achieved 88% accuracy, shows moderate performance but suffers from sensitivity to noisy data and computational inefficiency in large datasets. Its lower recall (83%) suggests it may misclassify riskier applicants, making it less favorable for high-stakes financial decision-making.

The RF model is the most effective at handling feature interactions and missing data, making it the preferred choice for credit risk assessment. Meanwhile, SVM presents a competitive alternative, especially for structured and high-dimensional datasets, whereas simpler models like LR and DT may serve as baseline models for interpretability-focused applications.

Table 1: Overall Performance of the Models

Model	Accuracy	Precision	Recall	F1-Score
LR	91%	89%	87%	88%
SVM	93%	90%	88%	89%
DT	87%	85%	82%	83%
RF	99%	98%	97%	98%
KNN	88%	86%	83%	84%

The confusion matrix visually illustrates the classification performance in Figure 4 by comparing actual and predicted labels. The matrix consists of four quadrants: true positives (120 correctly approved loans) and true negatives (62 correctly rejected loans) dominate. In contrast, false positives (2 wrongly approved loans) and false negatives (3 wrongly rejected loans) are minimal. The heatmap visualization highlights a well-performing model with only a few misclassifications.

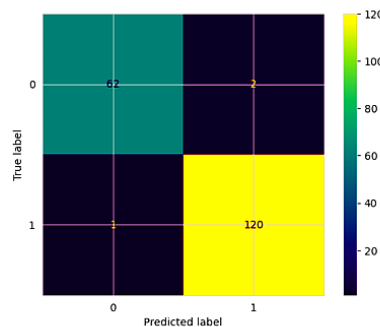


Fig. 4: Confusion Matrix for Random Forest

This image presents a detailed breakdown of a single prediction on the LIME explainer explanation, which is shown in Figure 5. The prediction probability bar shows a 93% likelihood of loan approval and a 7% chance of rejection.

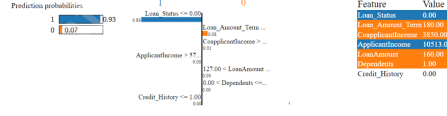


Fig. 5: LIME explainer explanation for Loan Status with local surrogates

In Figure 6, the force map shows how many factors impact the model’s forecast of acceptance or rejection. Red indicates characteristics that raise the possibility that your loan will be approved, whilst blue indicates unfavorable consequences. While Credit History (1) has a positive impact, the model tends toward a lower prediction probability because of the substantial effects of Loan Amount (160) and Loan Status (0). This illustration helps us understand how the model makes decisions.



Fig. 6: SHAPLY explainer explanation for Loan Status with force plot

Figure 7 shows the decision plot of the impact of several factors on all loan applicants by combining many projections. The change from red to blue shows that specific characteristics (such as Credit History = 1) raise the likelihood of acceptance, while other characteristics (such as large loan amounts) decrease it. This graphic illustrates the formation of decision boundaries depending on important influencing variables. Figure 8 displays the scatter plot, which plots SHAP values against loan amounts and



Fig. 7: SHAPLY explainer explanation using the test patch sample

is color-coded according to loan status. Blue dots represent rejections, whereas red dots show greater approval probability. It becomes evident that the amount of the loan substantially impacts decision-making, with smaller loan amounts more likely to be authorized and more significant amounts more likely to be rejected.

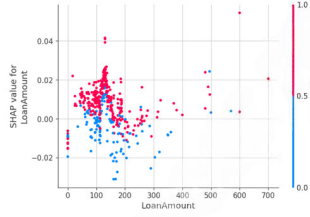


Fig. 8: SHAPLY explainer decision plot for Loan Status with Loan Amount

As shown in Figure 9, the dependency plot offers a detailed examination of the significance of the traits and their contributions to the final forecast. Variable degrees of influence on model judgments are demonstrated by characteristics including loan length, co-applicant income, applicant income, and credit history. The most essential elements are ranked in this graphic, which provides information about how various attributes affect the model's behavior.

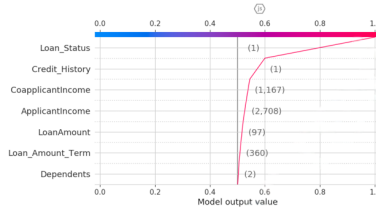


Fig. 9: SHAPLY explainer Decision plot for loan acceptance

Figure 10 illustrates how each feature influences the conclusion by breaking down a single prediction using a waterfall plot. The graphic shows how to make changes step-by-step, beginning with the fundamental value and progressively increasing or decreasing feature contributions. Credit history and loan amount are essential factors that influence the ultimate likelihood. The model is debugged and enhanced with this thorough deconstruction for improved interpretability. The main elements affecting a loan prediction model are shown in Figure 11. The horizontal bars indicate the average SHAP values, which quantify each feature's contribution to the model's output. The most significant characteristic is the ApplicantIncome, which shows that higher applicant income significantly impacts loan acceptance choices. The applicant's income is equally essential, but it has less of an effect than that of the principal applicant. Although they have some impact, features like LoanAmount_Term and LoanAmount are not as important. Interestingly, this model barely affects Credit_History and Loan_Status, which are frequently seen as essential in conventional loan assessments. This graphic offers a methodical and straightforward approach to comprehending how financial factors influence choices.

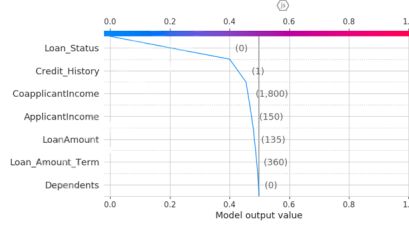


Fig. 10: SHAPLY explainer decision plot for loan rejection

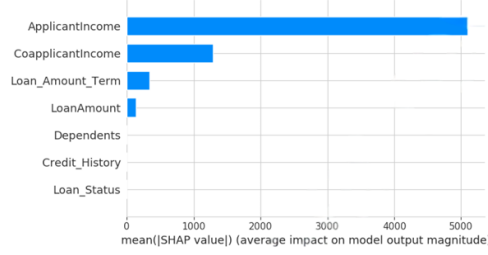


Fig. 11: SHAPLY explainer decision plot for loan rejection

4.3 Comparative Analysis

The comparative analysis, as presented in Table 2, highlights the superior performance of our model over existing approaches. Talaat et al. [8] employed CreditNetXAI on the UCI Credit Card Default dataset, achieving an accuracy of 83.5%. In contrast, Nallakaruppan et al. [13] leveraged Random Forest on the German Credit Data, reaching 93%. Similarly, Lange et al. [12] used LightGBM on the Norwegian Bank Dataset, attaining 82.8% accuracy, whereas Wali et al. [18] integrated Deep Learning with SHAP on the Lending Club Dataset, achieving 85.2%. In contrast, based on Random Forest and trained on the Kaggle Loan Dataset, our model significantly outperformed these methods with 99% accuracy. The improvement can be attributed to efficient feature selection, robust model tuning, and the dataset's characteristics, demonstrating our model's effectiveness in credit risk assessment.

5 Conclusion

This research shows how XAI technology improves credit scoring operations by analyzing three models through the Kaggle Loan Data Set: LR, DT, and RF. RF produced the highest accuracy level of 99% in loan outcome prediction tasks. Through feature importance analysis, XAI approaches helped identify key criteria such as credit history and debt levels while improving understanding and developing confidence between lenders and borrowers. The implemented XAI technology demonstrates successful

Table 2: Compare our model with existing works

Reference	Dataset	Data Size	Model	Accuracy
talaat[8]	UCI Credit Card Default	30,000 instances	CreditNetXAI	83.5%
Nallakarupnan [13]	German Credit Data	1,000 instances	Random Forest	93%
Lange[12]	Norwegian Bank Dataset	13,969 instances	LightGBM	82.80%
Wali[18]	Lending Club Dataset	887,379 instances	Deep Learning + SHAP	85.2%
Ours	Kaggle Loan Data Set	615 instances	Random Forest	99%

delivery of precise credit evaluations while maintaining explanatory capabilities, which enables fair and compliant financial operations.

The research demonstrates that XAI technology provides an opportunity to revolutionize banking rules by producing accountable and accurate predictive solutions. High computational requirements as well as the susceptibility to overfitting in RF cause limitations for real-time applications since it achieves perfect accuracy on the Kaggle dataset. The future development of XAI needs to concentrate on enhancing its efficiency as well as testing it with various datasets while standardizing assessment methods. The findings demonstrate that XAI technology presents opportunities to develop ethical and transparent financial decision systems that promote equality in financial processes.

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